PSG COLLEGE OF TECHNOLOGY
CODING CLUB

Presents

AI4Dev ‘26 Hackathon

A national-level AI-Enabled Transformative Technologies

for Global Development Hackathon

Team ID: PS040098

PROBLEM DEFINITION

Domain: Healthcare and life sciences

Project Title : AI System for Detecting Diabetic Retinopathy from Eye Images.

Problem Statement :

Diabetic Retinopathy damages the blood vessels in the retina and is one of the major causes of blindness in diabetic patients. Early stages often show very few symptoms, so many patients are diagnosed only after the disease becomes severe. Manual screening of retinal images requires skilled ophthalmologists and is time consuming, especially in areas where medical specialists are limited

Objective:

The objective of this project is to develop an Artificial Intelligence based system that can analyze retinal images and detect early signs of diabetic retinopathy. By using image processing and deep learning techniques, the system aims to support doctors in faster screening, improve early diagnosis, and help reduce the risk of blindness caused by diabetes.

EXISTING SOLUTION AND GAP

Existing Solution

- Screening is mainly performed by ophthalmologists
- Retinal images are captured using fundus cameras
- Doctors manually examine images to detect abnormalities
- Screening programs are conducted in hospitals and clinics
- Telemedicine allows retinal images to be sent to specialists
- Some hospitals use computer-aided diagnostic tools

Gap in Existing Solution

- Shortage of ophthalmologists in many regions
- Manual diagnosis is time-consuming
- Many patients cannot be screened regularly
- Diagnostic accuracy may vary between doctors
- Rural areas lack proper screening facilities
- Need for faster and more accessible AI-based detection systems

SOLUTION OVERVIEW

Solution Overview

- AI system detects Diabetic Retinopathy from retinal images
- Images captured using a fundus camera and preprocessed to improve quality
- Deep learning model analyzes images and classifies as normal, mild, moderate, or severe

Technology Stack

- Python
- TensorFlow & Keras
- OpenCV
- CNN model
- Web/Mobile integration

Bottlenecks

- Need large retinal image datasets
- Image quality variations may affect accuracy
- Reliable results required for doctors
- Patient data privacy

PROOF-OF-CONCEPT FOR MVP

Demo of the Solution:

The demo shows how the AI system detects Diabetic Retinopathy from retinal images. A retinal image is uploaded into the system as input. The image is preprocessed to improve clarity and highlight retinal features. The deep learning model then analyzes the image and identifies disease patterns. After processing, the system displays the prediction result indicating whether the retina is normal or shows signs of diabetic retinopathy. The output may also show the severity level, helping doctors quickly understand the condition and take further medical action.

Proof of Concept (PoC):

The Proof of Concept demonstrates the technical feasibility of the proposed system. A sample dataset of retinal images is used to train the deep learning model. The trained model is tested with new retinal images to evaluate its prediction capability. The system successfully classifies images as normal or diseased based on learned patterns, showing that Artificial Intelligence can assist early screening of diabetic retinopathy and be developed into a real-world healthcare application.

REALNESS

Originality: The proposed solution is an AI-based system designed to detect Diabetic Retinopathy from retinal images. It uses deep learning and image processing techniques to enable automated screening of retinal images. The system focuses on faster and early detection of the disease, which can support doctors in diagnosis and improve eye screening accessibility, especially in rural or underserved areas.

Feasibility: The system is feasible because it uses widely available technologies such as Python, TensorFlow, and OpenCV. Public retinal image datasets are available for training the deep learning model. Convolutional Neural Network (CNN) models are proven to be effective in analyzing medical images. Therefore, the system can be developed, trained, and tested in an academic or research environment.

Viability: The solution is viable as it can help hospitals screen large numbers of patients more efficiently. It reduces the workload of ophthalmologists by assisting in the initial detection process. Early diagnosis can lead to faster treatment and better patient outcomes. The system can also be integrated into web or mobile healthcare platforms for wider accessibility.

RESEARCH PAPERS REVIEWED (LITERATUREREVIEW DONE)

1. https://www.ijert.org/detection-of-diabetic-retinopathy-using-deep-learning](https://www.ijert.org/detection-of-diabetic-retinopathy-using-deep-learning

2.https://ijisae.org/index.php/IJISAE/article/view/5732

• Literature review of these research papers was conducted to understand existing methods and models used for diabetic retinopathy detection.

BUSINESS MODEL

Impact of the Solution:
The AI system enables early detection of Diabetic Retinopathy, helping prevent vision loss. It supports doctors by allowing faster retinal screening and improving early diagnosis. The solution is especially useful in rural areas where ophthalmologists are limited, helping patients receive timely treatment.

Market Opportunities:
The growing number of diabetes patients increases the demand for retinal screening. Hospitals, eye clinics, and diagnostic centers can adopt the system. It also has opportunities in telemedicine platforms and government rural health screening programs.

Business Model:
The system can follow a Software as a Service model for hospitals and clinics through monthly or yearly subscriptions. Another option is a per-scan pricing model, along with partnerships with hospitals and diagnostic centers.

Expected Cost of the Solution:
Existing retinal screening systems cost around ₹2,00,000–₹10,00,000. The proposed AI software system costs about ₹40,000–₹80,000, making it more affordable and scalable for small hospitals and rural clinics.

SUBMISSIONS

- Github public repository URL-

https://github.com/chirumerlasrilikitha/Diabetic-Retinopathy-AI-Detection/upload/main

- Video Presentation URL-

https://youtu.be/PsdYiord9Ac?si=KVKXq0MPVHarKwkp

THANK YOU